Constants

Symbol	Value	Unit
R	2.25	μm
a_{nom}	1.4706×10^{-2}	μm/pN
k_B	1.3806503×10^{-5}	pN·μm/K
T	310.15	K
$4k_BT$	1.7128×10^{-2}	pN·μm

Symbols and units

Symbol	Meaning	Unit
$x[k]$	position along axis	μm
$\bar{x} := x/R$	normalized position	–
$a_x[k]$	motion gain (mobility)	μm/pN
$a'_x := da_x/dx$	gain sensitivity	1/pN
$K_x(\bar{x})$	wall sensitivity of c	–
$f_T[k]$	thermal force	pN
$\sigma_x^2[k]$	single-step thermal position var	μm^2
$a_{det}[k]$	deterministic part of a_x	μm/pN
$a_{ram}[k]$	random part of a_x	μm/pN
$x_{ram}[k]$	position residual	μm

Wall model

$$a_x[k] = \frac{a_{nom}}{c(\bar{x}[k])}$$

Wall sensitivity

$$K_x(\bar{x}) := \frac{1}{c}\frac{dc}{d\bar{x}}, \qquad a'_x[k] := \frac{da_x}{dx} = -\frac{a_x[k]\, K_x(\bar{x}[k])}{R}$$

Thermal force

$$\boxed{\mathrm{Var}\big(f_T[k]\big) = \frac{4\,k_BT}{a_x[k]}}$$

Thermal-driven position variance (single step)

$$\boxed{\sigma_x^2[k] := a_x[k]^2 \cdot \mathrm{Var}\big(f_T[k]\big) = 4\,k_B T\,a_x[k]}$$

Gain decomposition

$$a_x[k] = a_{det}[k] + a_{ram}[k], \qquad a_{det}[m] = \frac{a_{nom}}{c(\bar{x}_d[m])}$$

$$a_{ram}[k] \approx a_x'[k] \cdot x_{ram}[k], \qquad x_{ram}[k] := x[k] - x_d[k]$$

Variance propagation

$$\boxed{\begin{aligned} \mathrm{Var}\big(a_{ram}[k]\big) &= a_x'^2[k] \cdot \mathrm{Var}\big(x_{ram}[k]\big) \\ \mathrm{Var}\big(\Delta a_{ram}[k]\big) &= a_x'^2[k] \cdot \mathrm{Var}\big(\Delta x_{ram}[k]\big) \end{aligned}}$$

Deterministic increment

$$\Delta a_x[k] = a_{det}[k+1] - a_{det}[k] = a_x'[k] \cdot \Delta x_d[k]$$